



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

General Metric Space

Properties of Metric, Modified Metric, NS, IPS

FEBRUARY 03, 2026

CONDUCTED BY

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

✉ partho.dhor@bracu.ac.bd | ✉ parthosutradhor@gmail.com

For updates subscribe on  [@ParthoSutraDhor](#)

Motivation of Metric

Metric Space

Metric Space

A **metric space** is a pair (X, d) where $X \neq \emptyset$ and $d : X \times X \rightarrow \mathbb{R}$ satisfies, for all $x, y, z \in X$:

$$(M1) \quad d(x, y) \geq 0;$$

$$(M2) \quad d(x, y) = 0 \iff x = y;$$

$$(M3) \quad d(x, y) = d(y, x);$$

$$(M4) \quad d(x, y) \leq d(x, z) + d(z, y).$$

Here d is called a **metric** on X .

Inequalities in Metric Spaces

Reverse Triangle Inequality

Let (X, d) be a metric space. Show that for all $x, y, z \in X$,

$$|d(x, z) - d(y, z)| \leq d(x, y).$$



Two-point inequality for distances

Let (X, d) be a metric space. Show that for all $x, y, z, w \in X$,



$$|d(x, y) - d(z, w)| \leq d(x, z) + d(y, w).$$

Discrete Metric Space

Discrete Metric

Let X be a non-empty set. Define $d : X \times X \rightarrow \mathbb{R}$ as

$$d(x, y) = \begin{cases} 0, & \text{if } x = y; \\ 1, & \text{if } x \neq y. \end{cases}$$

Show that d is a metric on X .



Modified Metrics

💡 Metric via a Fixed Point

Let (X, d) be a metric space and let $x_0 \in X$. Define a function $\delta : X \times X \rightarrow \mathbb{R}$ by

$$\delta(x, y) = \begin{cases} d(x, x_0) + d(x_0, y), & \text{if } x \neq y, \\ 0, & \text{if } x = y. \end{cases}$$

Show whether δ is a metric on X or not.

Bounded Transform of a Metric

Let (X, d) be a metric space. Define $d' : X \times X \rightarrow \mathbb{R}$ as

$$d'(x, y) = \frac{d(x, y)}{1 + d(x, y)}.$$

Show that d' is a metric on X .



Metric via Truncation

Let (X, d) be a metric space. Define $d' : X \times X \rightarrow \mathbb{R}$ as

$$d'(x, y) = \min\{1, d(x, y)\}.$$

Show that d' is a metric on X .



Pullback Metric along an Injection

💡 Pullback metric along an Injection

Let (Y, d) be a metric space and let $\alpha : X \rightarrow Y$ be an injection. Define $d' : X \times X \rightarrow \mathbb{R}$ as

$$d'(x, y) = d(\alpha(x), \alpha(y)).$$

Show that d' is a metric on X .

Isometric Spaces

Isometric Spaces

Let (X, d_X) and (Y, d_Y) be metric spaces. They are said to be **isometric** if there exists a bijection $\alpha : X \rightarrow Y$ such that for all $x_1, x_2 \in X$,

$$d_Y(\alpha(x_1), \alpha(x_2)) = d_X(x_1, x_2).$$

Here α is called an **isometry** from X to Y .

Motivation of Norm

Normed Space

Normed Space

A **normed space** is a pair $(X, \|\cdot\|)$ where X is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\|\cdot\| : X \rightarrow \mathbb{R}$ satisfies, for all $x, y \in X$ and $\alpha \in \mathbb{R}$ (or $\mathbb{C}$):

$$(N1) \quad \|x\| \geq 0;$$

$$(N2) \quad \|x\| = 0 \iff x = 0;$$

$$(N3) \quad \|\alpha x\| = |\alpha| \|x\|;$$

$$(N4) \quad \|x + y\| \leq \|x\| + \|y\|.$$

Here $\|\cdot\|$ is called a **norm** on X .

Metric Induced by a Norm

Let $(X, \|\cdot\|)$ be a normed space. Define $d : X \times X \rightarrow \mathbb{R}$ as

$$d(x, y) = \|x - y\|.$$

Show that d is a metric on X .

Motivation of Inner Product

Inner Product Space

Inner Product Space

An **inner product space** is a pair $(X, \langle \cdot, \cdot \rangle)$ where X is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{R}$ (or $\mathbb{C}$) satisfies, for all $x, y, z \in X$ and $\alpha \in \mathbb{R}$ (or $\mathbb{C}$):

$$(IP1) \quad \langle x, x \rangle \geq 0;$$

$$(IP2) \quad \langle x, x \rangle = 0 \iff x = 0;$$

$$(IP3) \quad \langle x, y \rangle = \overline{\langle y, x \rangle};$$

$$(IP4) \quad \langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle.$$

Here $\langle \cdot, \cdot \rangle$ is called an **inner product** on X .

Norm Induced by an Inner Product

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Define $\| \cdot \| : X \rightarrow \mathbb{R}$ as 

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Show that $\| \cdot \|$ is a norm on X .

Metric Induced by an Inner Product

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Define $d : X \times X \rightarrow \mathbb{R}$ as 

$$d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle}.$$

Show that d is a metric on X .

💡 Cauchy-Schwarz Inequality

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Show that for all $x, y \in X$,

$$|\langle x, y \rangle| \leq \sqrt{\langle x, x \rangle} \sqrt{\langle y, y \rangle}.$$

or equivalently,

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

Thank You!

We'd love your questions and feedback.

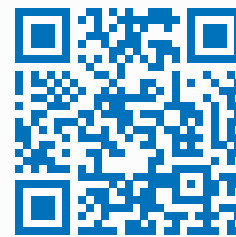
Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

✉ partho.dhor@bracu.ac.bd | ✉ parthosutradhor@gmail.com

 **@ParthoSutraDhor**

(Lectures, walkthroughs, and course updates)



Scan for the channel

References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.